Project: Amplitude Modulation and Demodulation Using BJT Amplifier and Diode Envelope Detector (DSB-FC)

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Abstract

— *This project presents the design and implementation of Double Sideband Full Carrier (DSB-FC) Amplitude Modulation and its demodulation using a diode-based envelope detector. A diode/BJT common-emitter amplifier is used for modulation, while a diode and RC low-pass filter are used for demodulation. The RC time constant is selected to satisfy $1/\omega_c \ll RC < 1/(\omega_m)$. Both stages are verified through MATLAB Simulink (ideal functional model) and Proteus (circuit-level simulation). The modulation index is varied to demonstrate under-modulation ($\mu < 1$), critical modulation ($\mu = 1$), and over-modulation ($\mu > 1$) conditions. Results confirm the fundamental principles of DSB-FC AM, the importance of proper RC time-constant selection, and the limitations of envelope detection under over-modulation.*

Index Terms — *Amplitude Modulation (AM), Double Sideband Full Carrier (DSB-FC), Envelope Detector, Modulation Index, RC Time Constant, BJT Common-Emitter Amplifier, MATLAB Simulink, Proteus*

I. INTRODUCTION

Amplitude Modulation (AM) is one of the foundational techniques of analog communications. In AM, the amplitude of a high-frequency *carrier signal* is varied in proportion to the instantaneous value of a lower-frequency *message signal*. This frequency translation permits efficient antenna design and frequency-division multiplexing of multiple channels. [1]

In this project, Double Sideband Full Carrier (DSB-FC) modulation is used, where the carrier and both sidebands are transmitted together. Although it is power inefficient (maximum efficiency ≈ 33.3%), it allows simple envelope detection at the receiver, making it suitable for low-cost broadcast radio systems. [1][2]

The receiver uses a diode envelope detector consisting of a half-wave rectifier and an RC low-pass filter. It does not require a synchronized carrier, which makes the receiver simple and low cost. This report includes the system design, key calculations, hardware implementation, and simulation results. [1]

A. Types of Amplitude Modulation:

The AM are of different types [1][2]:

Type	Carrier	Feature
DSB-FC	Full carrier transmitted	Simple envelope detection
DSB-SC	Carrier Suppressed	Better power efficiency
SSB-SC	Carrier Suppressed	Half bandwidth required
VSB	Partial carrier/ sideband	Used in TV broadcast

TABLE I: AM Modulation Variants

II. LITERATURE REVIEW

A. AM Signal Model:

Modulating signal: $m(t) = A_m \cos(2\pi f_m t)$

Carrier Signal: $c(t) = A_c \cos(2\pi f_c t)$

Modulated Signal: $\mathbf{s(t)=A_c[1+k_a \cdot m(t)]\cos(2\pi f_c t)}$ [1]

where ka is the amplitude sensitivity (V^{-1}) and the term "1" represents the DC offset added to ensure proper DSB-FC modulation.

For a single-tone message:

$$\begin{aligned} s(t) &= A_c \cos(2\pi f_c t) \\ &+ (\mu A_c/2) \cos[2\pi(f_c+f_m)t] \\ &+ (\mu A_c/2) \cos[2\pi(f_c-f_m)t] \end{aligned}$$

The three terms represent the **carrier** at fc, the **Upper Sideband (USB)** at fc + fm, and the **Lower Sideband (LSB)** at fc − fm. [1]

B. Bandwidth, Power and Efficiency:

Bandwidth for DSB-FC is $BW = 2\ fm$

Power in each component [2]:

$$P_c = A_c^2/2$$

$$P_{USB} = P_{LSB} = \mu^2 A_c^2/8$$

$$P_{total} = (A_c^2/2)(1 + \mu^2/2)$$

The power efficiency, since only sideband power carries information [1]:

$$\eta = \mu^2 / (2 + \mu^2)$$

At $\mu = 1$ (critical modulation), $\eta_{max} = 33.3$ %. The remaining 66.7 % is wasted in the carrier which is the fundamental power disadvantage of DSB-FC.

C. Modulation index:

The modulation index $\mu = k_a A_m / A_c$:

Modulation	μ	Effect on Envelope
Under	< 1	Envelope > 0 (always positive) ; no distortion
Critical	= 1	Envelope just touches zero
Over	> 1	Envelope distortion occurs; phase reversal and envelope detector fails

TABLE II: Modulation Index Regions [1]

III. OBJECTIVES

- Derive and analyze the mathematical model of DSB-FC AM: spectrum, bandwidth, power, efficiency, and modulation index.
- Implement AM modulation using (a) a diode-based nonlinear mixer and (b) a BJT CE variable-gain amplifier.
- Design and implement a diode envelope detector with analytically derived RC time constant.
- Simulate the full system in MATLAB Simulink and verify circuit-level behavior in Proteus.
- Analyze the effect of μ < 1, μ = 1, and μ > 1 on signal quality and demodulation fidelity.
- Identify and explain discrepancies between ideal and circuit-level simulation results

IV. METHODOLOGY

A. System Architecture:

The AM system comprises two subsystems: the **transmitter (modulator)** and the **receiver (demodulator)**.

The transmitter block diagram is shown in Fig. 1; the receiver block diagram in Fig. 2.

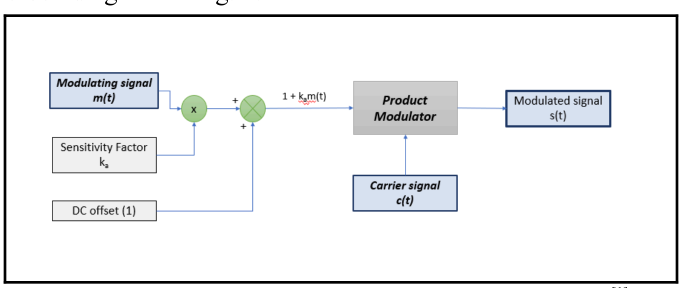

Fig. 1. Block diagram of AM transmitter (modulator side)[1]

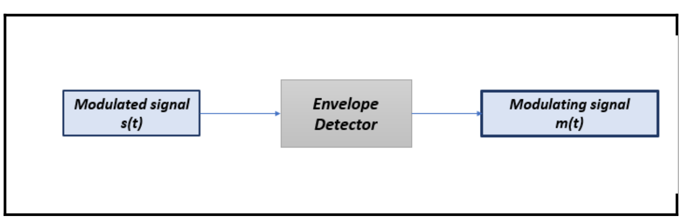

Fig. 2. Block diagram of AM receiver (demodulator side)[1]

B. Simulink Implementation:

In Simulink, a Sine Wave block generates a modulating signal m(t) and a second generates carrier signal c(t). An Add block computes [1 + ka·m(t)] and a Multiply block produces modulated signal s(t).

The envelope detector is implemented using an absolute value (Abs) block followed by a first-order low-pass filter (LPF).

Simulink Parameters are:

- `m(t) = Am sin(2π×5t)`
  `Am = 1 V, fm = 5 Hz`
- `c(t) = Ac sin(2π×50t)`
  `Ac = 1 V, fc = 50 Hz`
- k_a is varied to observe under-modulation, critical modulation, and over-modulation in the results section.
- Low pass filter has cutoff frequency = 10 and a time constant of 15.9 ms.

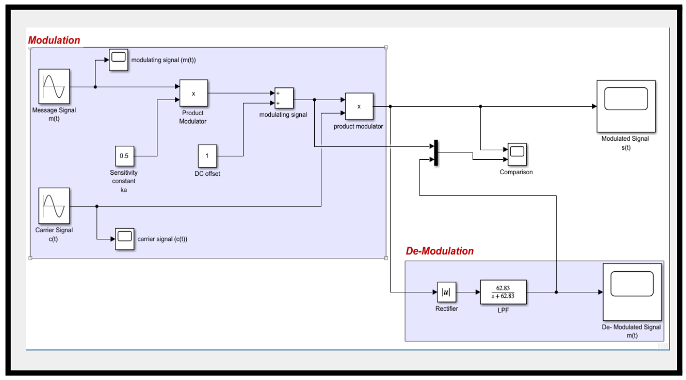

Fig. 3. Simulink implementation of AM modulator and envelope detector.

C. Diode Based Modulator:

In the modulation stage, the carrier and message signals are combined using resistors and then passed through a diode, which creates amplitude modulation by varying the carrier amplitude according to the message signal. An LC tank circuit is used to tune and filter the signal around the carrier frequency, removing unwanted components and producing a clean AM wave.

In the demodulation stage, a diode rectifies the AM signal, and an RC low-pass filter extracts the original message by following the signal's envelope.

Circuit Parameters are:

- Carrier frequency: ***fc = 150 kHz***
- Modulating frequency: ***fm = 500 Hz***
- Resistors = R1 = R2 = R3 = R4 = 10 kΩ
- Diode: D1(acts as non linear mixing element)
- LC tank: C1 = 1nF, L2 = 1mH
- Diode: D2 (rectifier diode)
- RC filter: R5 = 1kΩ, C2 = 470nF

Calculation:

- ***Resonant frequency = fr = 159.15 kHz***

The resonant frequency (**159.15 kHz**) is very close to carrier frequency (**150 kHz**). In practical analog circuits, this is close enough for the LC tank to successfully filter and pass the AM signal without squashing it.

- ***Time constant = RC = 470us***

Time period for carrier is 1/150,000 → 6.67 us

Time period for message signal is 1/500 → 2000 us
(1/fc) << time constant << (1/fm)
6.67 us << 470 us << 2000 us
The calculations satisfy the requirements for a clean demodulation.

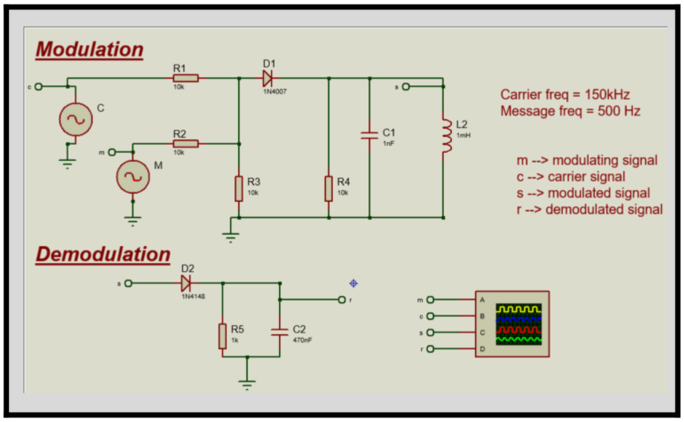

Fig. 4. Diode-based AM modulator circuit schematic (Proteus)

D. Transistor Based Modulator:

We used a BJT-based AM modulation and demodulation circuit. Unlike diode modulation, it uses a transistor (2N3904) which both amplifies and modulates the signal, making it closer to real transmitter systems.

In the modulation stage, the carrier signal is applied to the transistor base, and the message signal is applied to the emitter. The transistor amplifies the carrier, but its gain changes according to the message signal. This causes the carrier amplitude to vary, producing an AM signal at the collector output. The capacitor C3 helps by allowing the high-frequency carrier to pass while blocking the low-frequency message from interfering.

At the receiver, the diode removes the negative part of the signal then the RC filter smooths the waveform. Final output becomes the original 1 kHz message signal.

Circuit Parameters are:

- Carrier frequency: ***fc = 40 kHz***
- Modulating frequency: ***fm = 1 kHz***
- Transistor: Q1 = 2N3904 (NPN BJT acting as an amplifier)
- Biasing Resistors: R1 = 22 kΩ, R2 = 10 kΩ (Sets DC operating point)
- Amplifier Resistors: Re = Rc = 1 kΩ
- Capacitors: C1 = 10nF, C5 = 10uF, C3 = 100nF
- Diode: D1 (Rectifier)
- RC Filter: R6 = 10 kΩ, C4 = 22nF
- Coupling/Load: C2 = 10nF, R5 = 1 kΩ

Calculation:

- ***Time constant = RC = 220us***

Time period for carrier is 1/40,000 → 25 us
Time period for message signal is 1/1000 → 1000 us
(1/fc) << time constant << (1/fm)
25 us << 220 us << 1000 us
The calculations satisfy the requirements for a clean demodulation.

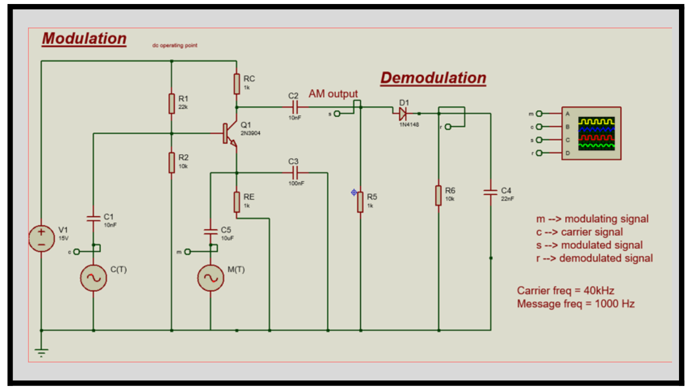

Fig. 5. Transistor based AM modulator circuit schematic (Proteus).

V. RESULTS & ANALYSIS

A. Simulink Results:

Modulating and Carrier signals are

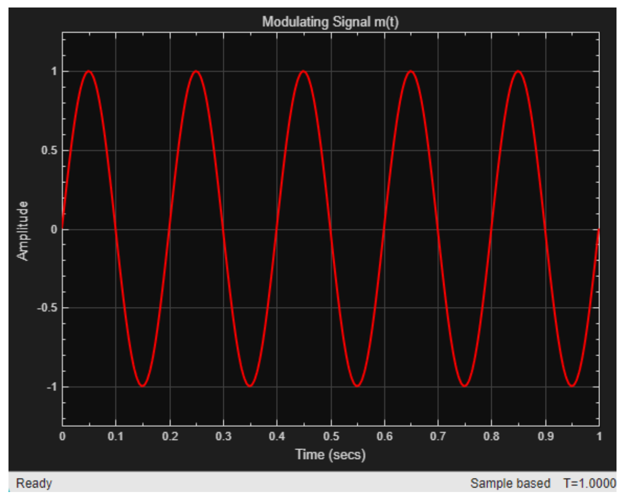

Fig. 6. Modulating signal m(t) = sin(2π×5t)

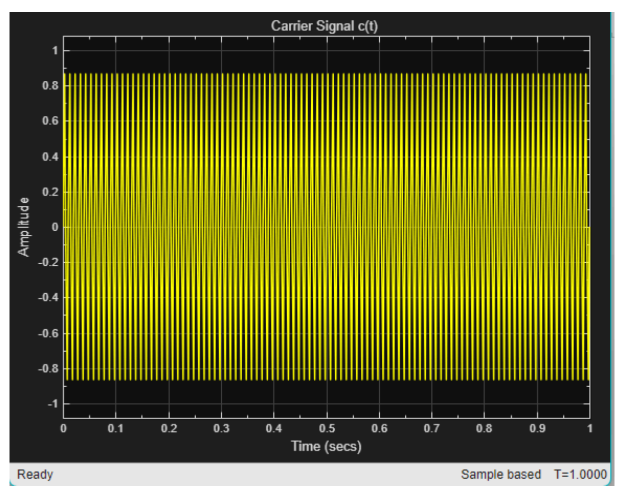

Fig. 7. Carrier signal c(t) = sin(2π×50t)

Three modulation conditions were simulated in Simulink by varying ka as Am = Ac =1 so μ = ka:

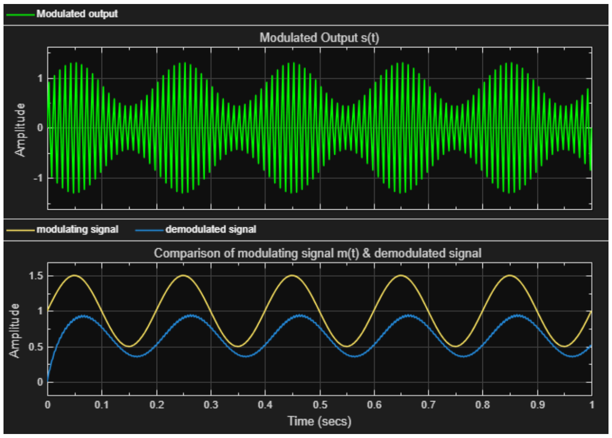

Fig. 8. Simulink: Under-modulation (ka = 0.5, μ = 0.5)

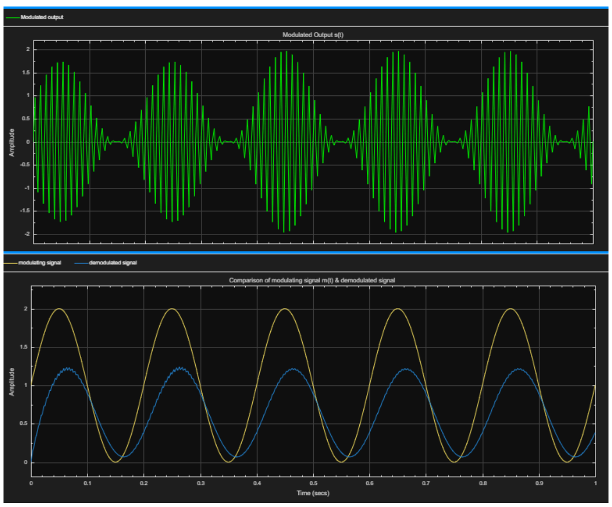

Fig. 9. Simulink: Critical modulation (ka = 1.0, μ = 1.0)

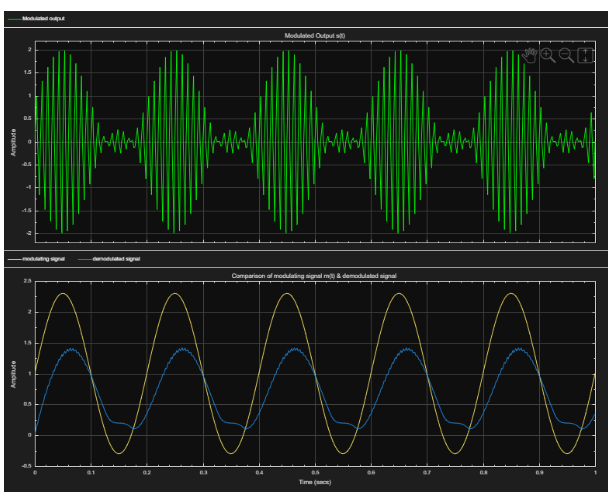

Fig. 10. Simulink: Over-modulation (ka = 1.3, μ = 1.3)

Analysis

Modulation	μ	Observation
Under	0.5	Clean envelope; demodulated output closely follows m(t) with slight smoothing due to the low-pass filter
Critical	1.0	Envelope just touches zero; correct recovery but slight distortion near zero-crossing points and reduced amplitude
Over	1.3	Envelope distortion with phase reversal, envelope detector fails to track correctly resulting in clipped and distorted output with loss of information

TABLE IV: Simulink Simulation Analysis

B. Diode based Modulator circuit Results:

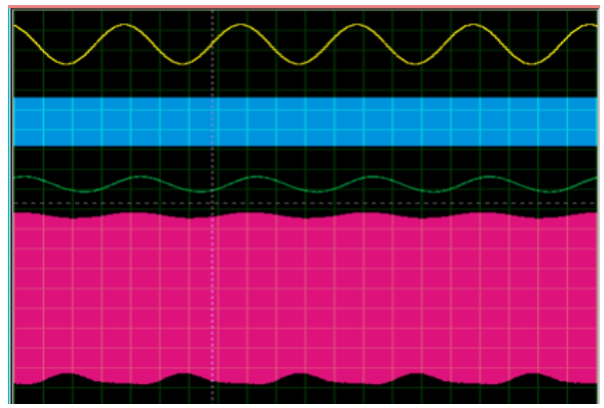

Fig. 11. Under-modulation (Am=2V, Ac=5V, μ=0.4).

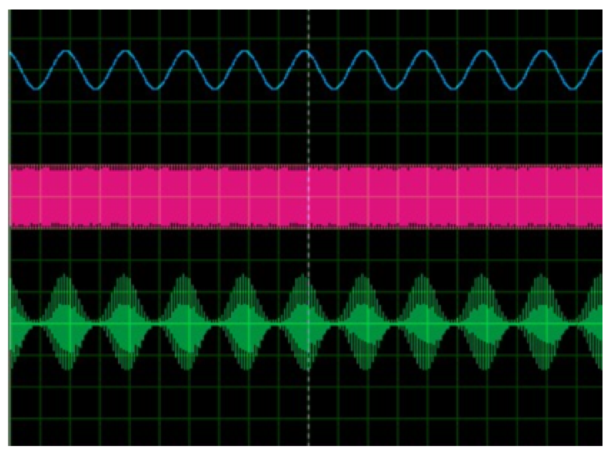

Fig. 12. Critical modulation (Am=Ac=1V, μ=1).

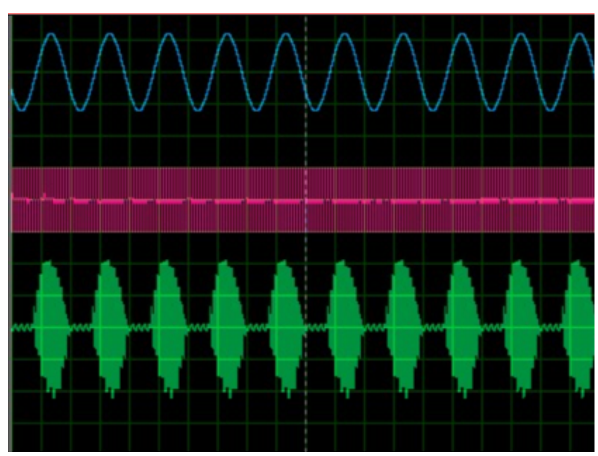

Fig. 13. Over-modulation (Am=5V, Ac=1V, μ=5)

Analysis

Modulation	μ	Observation
Under	0.4	Clean envelope; demodulated output closely follows m(t) with slight smoothing due to the low-pass filter
Critical	1	Very susceptible to noise
Over	5	Rectifier truncates the negative envelope leaving the clipped demodulated wave

TABLE V: Diode modulator circuit Analysis

C. Transistor based Modulator circuit Results:

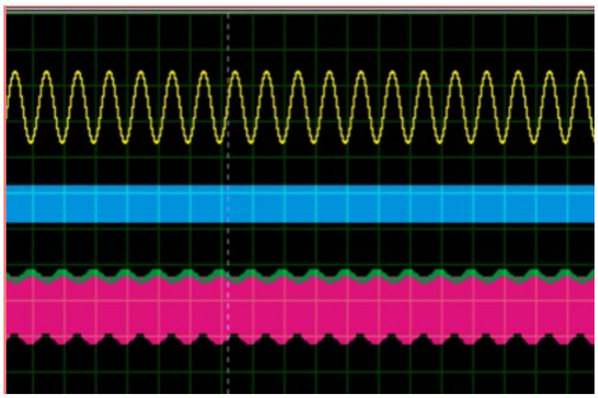

Fig. 14. Under-modulation (Am=2V, Ac=5V, μ=0.4).

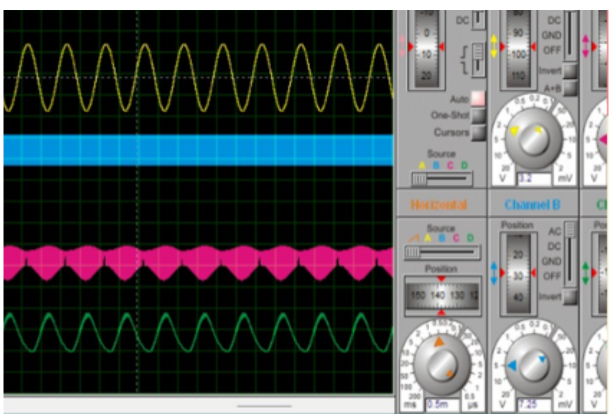

Fig. 15. Critical modulation (Am=Ac=5V, μ=1).

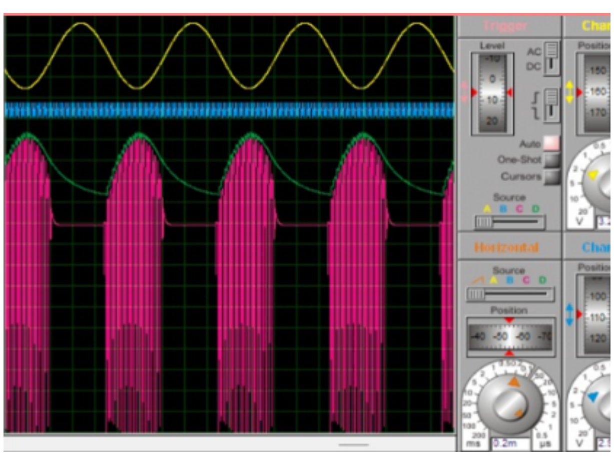

Fig. 16. Over-modulation (Am=5V, Ac=1V, μ=5).

Analysis

Modulation	μ	Observation
Under	0.4	Clean envelop, recovered m(t)
Critical	1	Pinched envelop, recovered m(t)
Over	5	Large phase reversal effect

TABLE VI: Transistor modulator circuit Analysis

D. Discrepancy Analysis:

Observed differences between Simulink and Proteus arise from:

- ***Parasitic Effects:*** Stray capacitances at 500 kHz cause phase errors and amplitude roll-off absent in the ideal Simulink model.
- ***BJT Nonlinearity:*** The exponential IC–VBE curve generates harmonics not present in the linear Simulink multiplier.
- ***Common Emitter (CE) Phase Inversion***: 180° output inversion in Proteus does not affect envelope detection but is visually evident in waveforms.
- ***Diode Forward Drop:*** ~0.7 V threshold in Proteus clips small-amplitude components; absent in ideal Simulink diode model.

E. Comparison of Diode and BJT based modulator:

Feature	Diode-Based Modulator	BJT-Based Modulator
Principle	Uses diode non-linearity for mixing	Uses transistor gain variation for modulation
Gain	No amplification	Provides signal amplification
Complexity	Simple circuit	Slightly more complex as biasing required`
Efficiency	Lower	Higher due to active device
Practical use	Basic AM demonstrations	Practical transmitters and RF systems
Signal quality	More distortion possible	Better linearity and control

We can conclude that The BJT-based modulator is more suitable for practical applications due to its amplification capability and better control over modulation compared to the diode-based approach.

VI. ADVANTAGES AND LIMITATIONS

Advantages:

- Simple receiver: Non-coherent envelope detection; no carrier synchronization required.
- Robust to frequency offset: Envelope detector is insensitive to small carrier phase errors.
- Wide adoption: AM broadcast band (535–1605 kHz) is internationally standardized.
- Low receiver cost: Billions of compatible AM receivers exist worldwide.

Limitations:

- Low efficiency: η_{max} = 33.3 % at μ = 1; two-thirds of transmitted power wasted in the carrier. [1]
- Amplitude noise susceptibility: Information in amplitude envelope is directly corrupted by additive noise.
- Bandwidth inefficiency: DSB-FC uses 2× the bandwidth of SSB-SC.
- Over-modulation risk: μ > 1 causes catastrophic envelope distortion and demodulation failure.

VII. APPLICATIONS

- AM Broadcast Radio (535–1605 kHz): Mass-market audio broadcasting using DSB-FC.
- Aviation VHF (118–137 MHz): Air-to-ground voice using AM for robustness and squelching.
- CB Radio: Short-range personal communications using AM and SSB-SC.
- Analog Television (VSB): Video signal broadcast using vestigial-sideband AM.
- QAM in Digital Systems: 2D generalization of AM used in Wi-Fi (up to 1024-QAM).
- HF RFID (13.56 MHz): ASK (digital AM) for data transmission in RFID tags.
- Fiber-Optic Systems: Intensity modulation of laser diodes (optical AM analogue).

VIII. CONCLUSION

This project successfully designed, analyzed, and simulated a complete DSB-FC AM communication system using both ideal (Simulink) and circuit-level (Proteus) platforms. The theoretical foundations — AM signal model, frequency spectrum with USB/LSB, bandwidth (BW = 2fm), power distribution, efficiency ($\eta = \mu^2/(2+\mu^2)$), and modulation index effects — were derived analytically. [1][2]

The BJT CE amplifier was demonstrated to be an effective variable-gain AM modulator through message injection at the emitter bypass node. The diode envelope detector was designed with τ = RC = 100 μs, satisfying the design constraint 0.318 μs ≪ 100 μs < 159.2 μs for fc = 500 kHz and fm = 1 kHz. [1][3]

All three modulation regimes were verified: under-modulation (μ < 1) produced clean recovery; critical modulation (μ = 1) produced marginal but acceptable recovery; over-modulation (μ > 1) caused irrecoverable envelope distortion. Discrepancies between Simulink and Proteus results were attributed to transistor nonlinearity, CE phase inversion, diode forward voltage drop, and parasitic effects.

IX. ACKNOWLEDGMENT

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IX. REFERENCES

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